

ISIC competition Deep Learning MHealth

kaggle.com/competitions/isic-competition-deep-learning-m-health/overview

Description

ISIC 2024 Competition Overview

The **International Skin Imaging Collaboration (ISIC)** has launched this competition to advance automated skin cancer detection by:

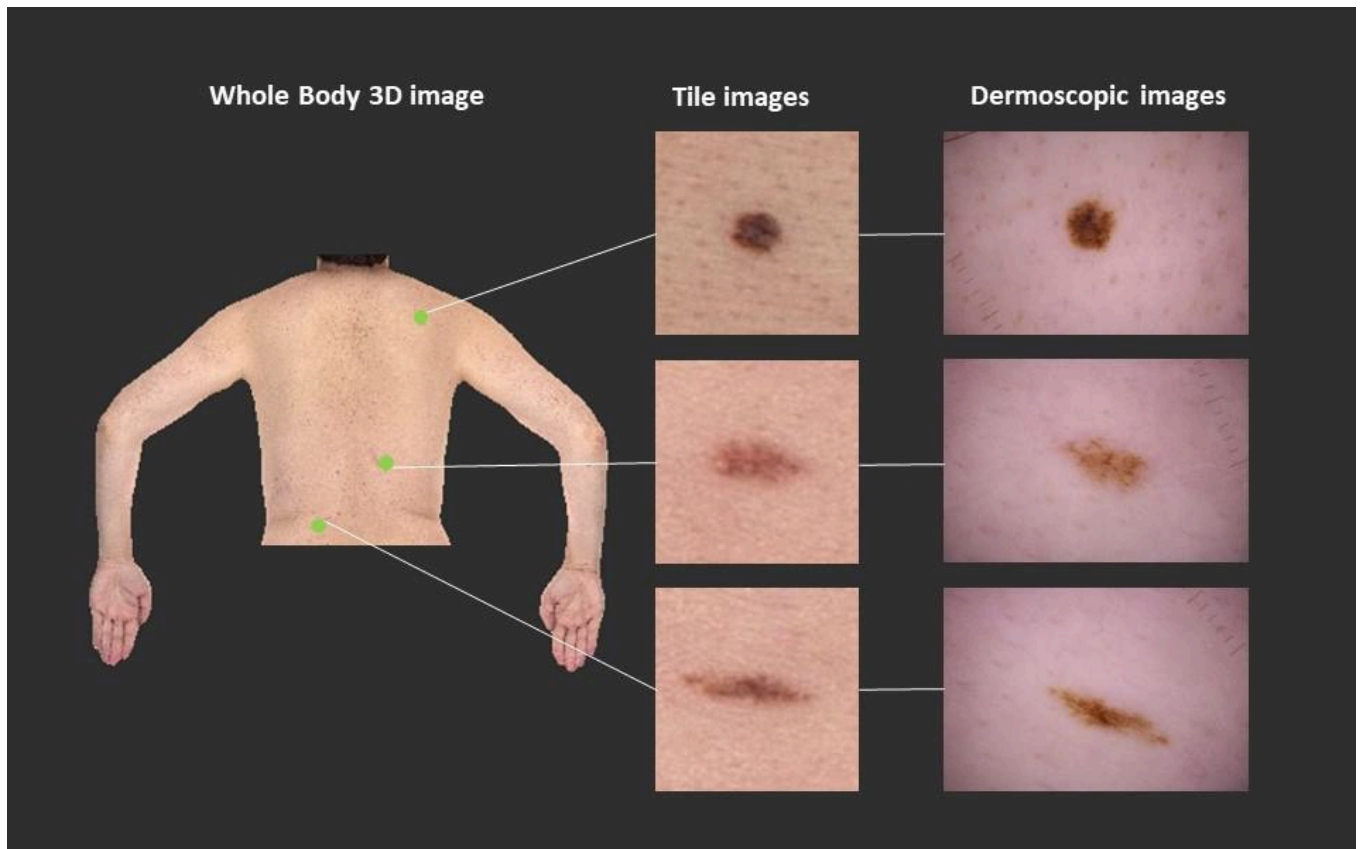
- **Improving accuracy** in distinguishing malignant from benign lesions
- **Enhancing efficiency** in clinical workflows
- **Developing algorithms** that prioritize high-risk lesions
- **Reducing mortality rates** by enabling earlier detection

Primary Task

You need to **classify skin lesions** as **benign** or **malignant**. For each lesion image (identified by `isic_id`), predict a **probability** in the range [0, 1] indicating the chance that the lesion is malignant.

High-Level Data Summary

- The dataset is called **SLICE-3D**, containing **skin lesion images** (JPEG files) cropped from 3D Total Body Photography (TBP).
- Each image has metadata in a corresponding `.csv` file, including:
 - **Binary diagnostic label** (`target` = 0 or 1)
 - **Patient data** (e.g., `age_approx`, `sex`, `anatom_site_general`)
 - **Additional attributes** (image source, diagnosis type)



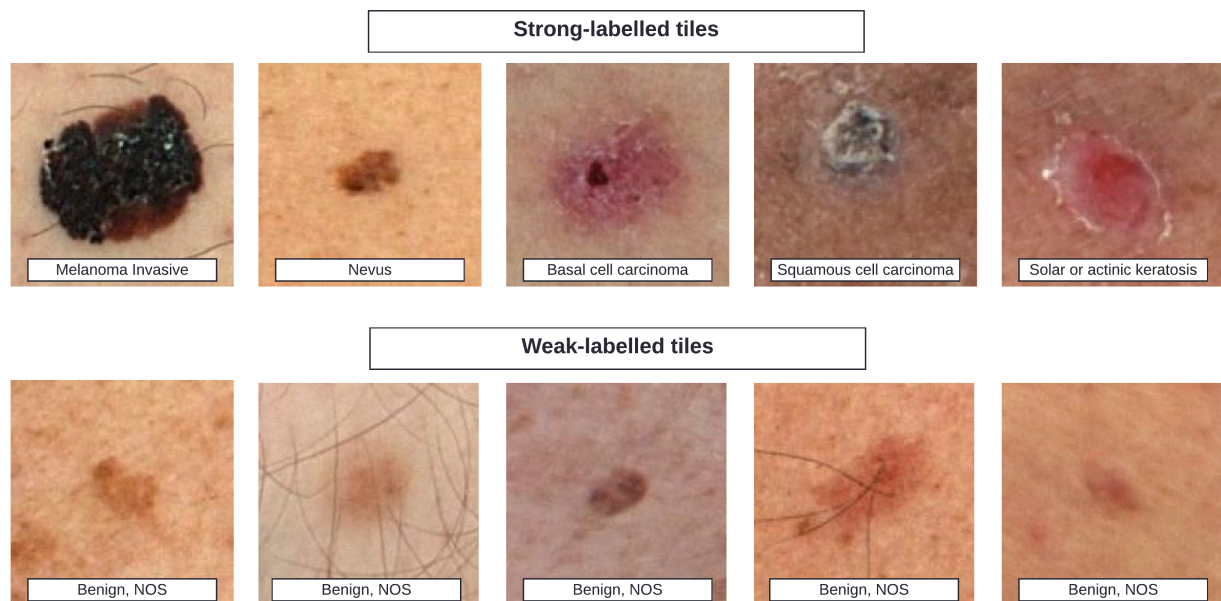
This challenge dataset mimics **non-dermoscopic images** using standardized 15x15 mm “tiles” of lesions from a 3D TBP system. Thousands of patients from multiple continents are represented, creating a broad, diverse dataset.

Task Description & Clinical Context

- **Why it matters:** Skin cancer can be deadly if not detected early. Many people lack access to dermatologic care, so accurate AI systems for image-based triage can improve outcomes.
- **Key goal:** Develop a binary classifier that identifies malignant lesions from a set of smartphone-quality images.
- **Impact:** This technology could help prioritize suspicious lesions (top K) for clinical review, especially in low-resource settings, potentially **saving lives** through earlier detection.

Importance of 3D TBP

The **3D Total Body Photography (TBP)** approach captures the entire skin surface in macro resolution. Each lesion on the patient’s body is automatically cropped as a 15x15 mm image tile. These images more closely resemble photos taken by a regular smartphone camera, as opposed to specialized dermoscopy devices.



Clinical Background

1. **Major skin cancer types:** Basal Cell Carcinoma (BCC), Squamous Cell Carcinoma (SCC), and Melanoma (most lethal).
2. **Early detection** is crucial: Minor surgery can cure many skin cancers if caught in time.
3. **Telemedicine implications:** With the rise in remote healthcare, patients often submit low-quality images captured at home. Robust AI models are needed to handle this variability.

Summary

- You will build a model to **classify skin lesions** (benign vs. malignant) with probabilities.
- The dataset includes **every lesion** from thousands of patients, reflecting real-world diversity.
- **3D TBP** and the “ugly duckling sign” concept illustrate the importance of comparing each lesion against the patient’s total lesion landscape.
- **Your work** can help improve early detection, prioritizing high-risk cases for clinical evaluation and potentially saving lives.

Evaluation

The evaluation section describes how submissions will be scored and how participants should format their submissions. You don't need a sub-title at the top; the page title appears above. Below is an example of a typical evaluation page.

Submissions are evaluated on [area under the ROC curve](#) between the predicted probability and the observed target.

Submission File

For each ID in the test set, you must predict a probability for the TARGET variable. The file should contain a header and have the following format:

```
isic_id,target ISIC_0015657,0.3 ISIC_0015729,0.3 ISIC_0015740,0.3
```